INCH-POUND
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SUPERSEDING
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27 August 1984

MILITARY SPECIFICATION

SWITCHES, ROTARY, ENCLOSED

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers enclosed snap action, spring return, positive detent, multipole and selection rotary switches.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-504 - Plastic Sheet and Film, Cellulose Acetate.
L-P-535 - Plastic Sheet (sheeting), Plastic Strip, Vinyl Chloride Polymer and Vinyl Chloride-Vinyl Acetate Copolymer, Rigid.
QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings and Structural and Special Shaped Sections.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, SEA 03Q42, Naval Sea Systems Command, 2531 Jefferson Davis Hwy, Arlington, VA 22242-5160 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-S-901	- Shock Tests, H.I. (High Impact), Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-E-917	- Electric Power Equipment, Basic Requirements (Naval Shipboard Use).
MIL-E-2036	- Enclosures for Electric and Electronic Equipment, Naval Shipboard.
MIL-L-3661	- Lampholder, Indicator Lights, Indicator Light Housing & Indicator Light Lenses, General Specification for.
MIL-S-8660	- Silicone Compound, NATO Code Number S-736.
MIL-P-15024	- Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5	- Plates, Identification.
MIL-P-15035	- Plastic Sheet, Laminated, Thermosetting, Cotton Fabric Base, Phenolic Resin.
MIL-P-15037	- Plastic Sheet, Laminated, Thermosetting, Glass Cloth, Melamine Resin.
MIL-F-15160	- Fuse, Instrument, Power, and Telephone, General Specification for.
MIL-S-15291	- Switches, Rotary, Snap Action and Detent/Spring Return Action, General Specification for.
MIL-T-16366	- Terminal, Electrical Lug and Conductor Splices, Crimp Style.
MIL-W-16878	- Wire, Electrical, Insulated General Specification for.
MIL-E-17555	- Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of.
MIL-F-19207	- Fuseholder, Extractor Post Type, Blown Fuse, Indicating and Nonindicating, General Specification for.
MIL-S-19622	- Stuffing Tube, Nylon, General Specification for.
MIL-F-21346	- Fuseholder, Block and Shroud Type, and Associated Fuse Clips, General Specification for.
MIL-S-21604	- Switches, Rotary, Multipole and Selector: General Specification for.
MIL-E-24142	- Enclosures for Electrical Fittings and Fixtures, General Specification for.

(See Supplement 1 for list of applicable specification sheets.)

FEDERAL

QQ-S-766	- Steel, Stainless and Heat Resisting, Alloys, Plate Sheet and Ship.

STANDARDS

FEDERAL

FED-STD-H28	- Screw Thread Standards for Federal Services.
FED-STD-H28/2	- Screw-Thread Standards for Federal Services, Section 2, Unified Thread Form and Thread Series for Bolts, Screws, Nuts, Tapped Holes and General Applications.

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MIL-STD-108 - Definition of and Basic Requirements for Enclosures for Electrical and Electronic Equipment.
MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MS28775 - Packing, Preformed, Hydraulic, Plus 275 Deg F, (O-Ring).

2.1.2 Other Government drawings. The following other Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWING

NAVAL SEA SYSTEMS COMMAND
NAVSHIP - 9000-S6202-73055 - Fabricated Terminals (Closed end) and Branch Buses for Electrical Fittings.

(Copies may be obtained from Commander, Portsmouth Naval Shipyard, Naval Engineering Drawing Support Activity, Code 202.2, Portsmouth, NH 03804-5000.)

(Unless otherwise indicated, copies of federal and military specifications, and standards are available from the Standardization Documents Order Desk, Buidling 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
ANSI B46.1 - Surface Texture. (DoD adopted)

(Application for copies should be addressed to American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
B 36/B 36M - Standard Specification for Plate Brass, Sheet, Strip and Rolled Bar.
B 121/B 121M - Standard Specification for Plate, Leaded Brass, Sheet, Strip, and Rolled Bar. (DoD adopted)
B 633 - Standard Specification for Zinc on Iron and Steel, Electrodeposited Coatings of. (DoD adopted)

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.3 Materials. Materials shall be as specified (see 3.1). When not specified, the manufacturer may select any material that will perform the intended function in the equipment and will otherwise comply with the requirements of this specification.

3.3.1 Metals. Metals shall be of a corrosion-resistant type or shall be treated to resist corrosion as specified in MIL-E-917. Dissimilar metals, in contact with each other shall be selected to comply with the requirements of MIL-E-917. The selection of metals in contact shall also be such as to prevent galling.

3.3.2 Nonmetallic materials. As specified in MIL-E-917, nonmetallic materials shall be nonflammable, nontoxic, and fungus-inert.

3.3.3 Prohibited materials. Asbestos, cadmium and mercury shall not be used in the manufacture of the enclosed switches. In addition, no parts shall have come in direct contact with metallic mercury, mercury compounds or mercury bearing instruments or devices, employing only a single boundary of containment, during manufacture or inspection. (A single boundary of containment is one which is not backed up by a second seal or barrier to prevent escape and consequent mercury contamination in the event of rupture of the primary seal or barrier.)

3.3.4 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

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3.4 Parts.

3.4.1 Switches. Unless otherwise specified (see 3.1), switches shall conform to MIL-S-15291 or MIL-S-21604.

3.4.2 Switch handles. Switch handles shall be of the materials and dimensions specified (see 3.1). Screws used to secure the handle to the shaft extension shall be corrosion resistant steel (CRES) commercial grade.

3.4.3 Switch shaft extensions and couplings. Switch shaft extensions and couplings shall be of the materials and dimensions specified (see 3.1). Pins used to secure the coupling to the switch shaft shall be nickel copper alloy in accordance with QQ-N-281. 302 stainless steel in accordance with QQ-S-766 may be used as an alternate.

3.4.4 Commercial (hardware) parts. Commercial utility hardware items such as screws, bolts, and nuts may be used, provided they possess properties equivalent to and are replaceable by MS or AN part numbers. Unless otherwise indicated (see 3.1), attaching hardware shall be as follows:

(a) Screws, nuts and bolts shall be brass commercial grade except they shall be CRES commercial grade, zinc-coated in aluminum enclosures.
(b) Lockwashers shall be phosphor bronze, commercial grade for brass hardware and CRES, commercial grade, zinc-coated for CRES hardware.
(c) Washers for switch handles shall be laminated plastic, type FBS in accordance with MIL-P-15035.

3.4.5 Fuses. Fuses shall conform to MIL-F-15160. The type shall be as specified (see 3.1). Unless otherwise specified (see 6.2), fuses shall not be furnished.

3.4.6 Fuse clips. Fuse clips shall conform to MIL-F-21346. Fuse clip retainers shall not be used with fuse clips for fuses rated 0-60 amperes. The type shall be as specified (see 3.1).

3.4.7 Fuseholders. Fuseholders shall conform to MIL-F-19207. The type shall be as specified (see 3.1).

3.4.8 Fuse base. The fuse base shall be of glass melamine conforming to type GME of MIL-P-15037 or shall be a material selected from laminates specified in MIL-E-917 provided they meet the electrical and mechanical requirements (see 3.1). The laminates shall be furnished in natural color.

3.4.9 Terminal assemblies. Where specified (see 3.1), terminal assemblies and buses shall be in accordance with Drawing 9000-S6202-73055.

3.4.10 Glands and bushings. Glands and bushings shall be of the same material as the basic enclosure.

3.4.11 Lamps. Lamps shall be of the incandescent type conforming to industry number 1769. Unless otherwise specified (see 6.2), lamps shall not be furnished.

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3.4.12 Light indicators and target cap assemblies. Light indicators and cap assemblies shall conform to MIL-L-3661. The style shall be as specified (see 3.1).

3.4.13 Cable clamps. When specified (see 3.1) commercial nylon cable clamps shall be furnished.

3.4.14 Wire terminals. When specified (see 3.1), wire terminals shall conform to MIL-T-16366.

3.4.15 Wire. When specified (see 3.1), wire shall conform to MIL-W-16878.

3.4.16 Preformed packing (O-rings). O-rings shall be in accordance with MS28775, of the sizes as specified (see 3.1), and shall be lubricated with silicon compound conforming to MIL-S-8660 at time of assembly.

3.4.17 Mounting parts. Unless otherwise specified (see 3.1), mounting parts such as angles, pads, brackets, straps and studs shall be of the same material as the basic enclosure.

3.5 Design and construction. Enclosed rotary switches shall be of the design, construction, and physical dimensions specified (see 3.1).

3.5.1 Enclosures. Enclosures shall be in accordance with MIL-E-24142 or shall be modified versions of MIL-E-24142 enclosures (see 3.1). After drilling holes in the enclosure, the openings shall be returned to the level of effectiveness of enclosure with replaceable seals. When specified (see 3.1), the seals shall be O-rings conforming to MS28775. When not specified (see 3.1), seals may be manufacturer's choice, provided the completed enclosure meets the applicable effectiveness of enclosure requirement.

3.5.1.1 Cable entrance. When specified (see 3.1), cable entrance into unit enclosures shall be by means of stuffing tubes conforming to MIL-S-19622. Unless otherwise specified (see 3.1 and 6.2), stuffing tubes, packing, and holes for stuffing tubes shall not be provided.

3.5.2 Switch orientation and mounting. Switch orientation within the enclosure shall be as specified (see 3.1). Switch mounting shall be such as to prevent switch movement on or lateral to the mounting centerline. For base mounted switches, balance pads secured to the enclosure may be used as required. Switches may be mounted on pads, brazed directly to the enclosure or on angles, straps, plates or a combination thereof. The actual method of mounting the switch within the enclosure shall be manufacturer's choice except that switches shall not be modified to suit mounting.

3.5.3 Wiring diagram. Where required (see 3.1), the wiring diagram shall be in accordance with MIL-E-2036, except the plastic protective material shall conform to L-P-504 or L-P-535, as applicable. Sufficient blank space shall be available for depicting the external switch connections after installation.

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3.5.4 Local wiring. Where required (see 3.1), local wiring shall be installed by the contractor.

3.5.5 Threaded parts and fastening devices. Unless otherwise specified (see 3.1), threads for all threaded parts and fastening devices shall be in accordance with MIL-E-917, except they shall conform to FED-STD-H28 and shall be right hand, coarse-threaded series, unified form, class 2A or 2B conforming to FED-STD-H28/2.

3.5.6 Welding and brazing. Welding and brazing of non-electrical parts shall be in accordance with commercial practices and shall otherwise meet all of the requirements specified herein.

3.5.6.1 Brazing. Parts or components shall be fastened by intermittent localized brazing in such a manner as to ensure that each enclosure assembly will not be completely soft-annealed and will not distort or leak when subjected to the shock tests (see 4.7.6).

3.5.7 Soldering. Soldering shall be in accordance with requirement 5 of MIL-STD-454.

3.5.8 Stress relief. All metals used in the switch enclosure assembly shall be treated or heat treated to prevent deterioration or failure due to stresses or other conditions resulting from working, forming, and similar processs during fabrication.

3.5.9 Drilling, countersinking, and tapping. Drilling, countersinking, and tapping shall be done before plating or finish is applied.

3.5.10 Electrical creepage and clearance distances. Electrical creepage and clearance distances shall be in accordance with MIL-E-917.

3.6 Performance requirements.

3.6.1 Dielectric withstanding voltage. For enclosed switches having electrical circuits other than switches (see 3.1), there shall be no evidence of disruptive discharge, arcing (other than at test probes), flashover, corona (visible, audible, or smell), or breakdown when tested as specified in 4.7.2.

3.6.2 Insulation resistance. For enclosed switches having electrical circuits other than switches (see 3.1), the insulation resistance of the electrical circuits shall be not less than 10 megohms when tested as specified in 4.7.3.

3.6.3 Mechanical operation. The enclosed rotary switches shall be mechanically operative and shall exhibit no sign of binding of the operating shaft during and after the test of 4.7.4.

3.6.4 Vibration. When enclosed rotary switches are tested as specified in 4.7.5, there shall be no opening of closed switch contacts greater than 10 microseconds and there shall be no change in switch shaft position or evidence of broken, loose, deformed, or displaced parts.

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3.6.5 Shock. When enclosed rotary switches are tested as specified in 4.7.6, there shall be no opening of closed switch contacts of 20 milliseconds and there shall be no change in switch shaft position, or evidence of broken, loose, deformed, or displaced parts. Following the test, the enclosed rotary switches shall pass the dielectric withstanding voltage tests (if applicable) and the level of effectiveness of enclosure (see 4.7.2 and 4.7.7).

3.6.6 Level of effectiveness of enclosure. The enclosed rotary switches shall meet the specific degree of enclosure requirements specified (see 3.1) in accordance with MIL-STD-108 when tested as specified in 4.7.7. Following the test, the enclosed rotary switches shall satisfactorily pass the dielectric withstanding voltage test (if applicable) (see 4.7.2).

3.6.7 Salt spray. When enclosed switches are tested as specified in 4.7.8, there shall be no evidence of excessive corrosion. Excessive corrosion is defined as that which interferes with electrical or mechanical performance, or in the case of plated metals, corrosion which has migrated through the plating and attacked the base metal. Where corrosion-resisting treatments are used, it will be permissible to have limited superficial corrosion such as would result from scratches.

3.7 Finish.

3.7.1 Surface finish. Surface finishes specified (see 3.1), shall be in accordance with ANSI B46.1.

3.7.2 Zinc coating. Zinc coating, where required (see 3.1), shall be in accordance with ASTM B 633.

3.7.3 Painting. Unless otherwise specified (see 3.1), brass enclosures shall be neither primed nor painted (see 6.2).

3.8 Marking.

3.8.1 Information and identification plates. Information and identification plates shall be attached to the box cover by means of screws in tapped holes or screws held by pads brazed to the back of the cover. Unless otherwise specified (see 3.1), plates for steel and brass boxes shall be brass and plates for aluminum shall be CRES types A, B, and C in accordance with MIL-P-15024 and severe service MIL-P-15024/5.

3.8.1.1 Identification plates. Marking information shall be as specified (see 3.1). Plate size shall be as specified (see 3.1). Where not specified rectangular plates shall be of the standard dimensions specified in MIL-P-15024.

3.8.1.2 Information plates. Marking shall be as specified (see 3.1 and 6.2.1). Where marking information is not specified, information plates shall be furnished blank. Plate size shall be as specified (see 3.1).

3.8.1.3 Caution plate. Caution plates shall be as specified (see 3.1).

3.9 Workmanship. Workmanship shall be in accordance with requirement 9 of MIL-STD-454.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The inspections specified herein are classified as follows:

(a) First article inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202.

4.4 First article inspection. First article inspection shall be performed on sample units which have been produced with equipment and procedures normally used in production (see 6.3).

4.4.1 Sample size. One enclosed switch of each symbol number shall be subjected to first article inspection.

4.4.2 Inspection routine. The sample shall be subjected to the inspections specified in table I.

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TABLE I. First article inspection.

Inspection	Requirement paragraph	Test method paragraph
Visual and dimensional examination	3.1, 3.3, 3.4, 3.5, 3.7, 3.8 and 3.9	4.7.1
Dielectric withstanding voltage	3.6.1	4.7.2
Insulation resistance (where applicable)	3.6.2	4.7.3
Mechanical operation	3.6.3	4.7.4
Vibration	3.6.4	4.7.5
Shock	3.6.5	4.7.6
Level of effectiveness of enclosure	3.6.6	4.7.7
Salt spray	3.6.7	4.7.8

4.4.3 Failures. One or more failures shall be cause for refusal to grant first article acceptance.

4.4.4 First article extension. First article acceptance for a given size enclosure with a single switch may be used as evidence of acceptance of smaller sizes (see 6.4.1). The extension, however, in no way changes the quality conformance inspection in accordance with 4.5.

4.5 Quality conformance inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspection.

4.5.1.1 Lot. For the purpose of quality conformance inspection and test sampling, a lot is defined as all the enclosed switches of the same military part number (symbol number), produced in one facility, using the same production processes and materials, and being offered for delivery at one time.

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4.5.1.2 Sampling for group A inspection. As a minimum, the contractor shall randomly select a sample quantity from each lot of completed enclosed switches in accordance with table III and inspect them in accordance with table II for the defects listed in table IV. If one or more defects are found in any sample, the entire lot represented by the sample shall be rejected. If a lot is rejected, the contractor has the option of screening 100 percent of lot for the defective characteristic(s) or providing a new lot which shall be inspected in accordance with the sampling plan contained herein. The contractor shall maintain for a period of 3 years after contract completion all records of inspections, tests, and any resulting rejections.

TABLE II. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph
Visual and dimensional examination	3.1, 3.3, 3.4, 3.5, 3.7, 3.8 and 3.9	4.7.1
Mechanical operation	3.6.3	4.7.4

TABLE III. Sampling for group A inspection.

Lot size	Sample size	
	Major defects	Minor defects
2 to 8	All	3
9 to 25	8	3
26 to 50	8	5
51 to 90	8	6
91 to 150	12	7
151 to 280	19	10
281 to 500	21	11
501 to 1,200	27	15
1,201 to 3,200	35	18
3,201 to 10,000	38	22

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TABLE IV. Classification of defects.

Categories	Defects
Critical:	None defined.
Major	
101	Symbol not as specified.
102	Parts missing, damaged, nonconforming, improperly fitted, or assembled.
103	Enclosure not as specified.
104	Light indicators/fuses not as specified; not mounted and oriented as specified; not marked as specified.
105	Switch not type specified; not mounted and oriented as specified; not marked as specified.
106	Dimensions not as specified.
107	Evidence of drilling, countersinking, and tapping after plating or application of finish.
108	Materials less than minimum requirements; evidence of unauthorized materials used; evidence of cracks, splits, seams, or pitting.
109	Threads nonconforming, form not as specified, not to size, missing, chipped, crossed, stripped, or damaged.
110	Terminal screws and lockwashers missing, damaged or nonconforming.
111	Sharp corners of components not rounded as required.
112	Plating, when specified nonconforming, missing, or incomplete; not smooth or uniform, evidence of peeling, nonadherent or uncoated areas; evidence of oxide scale or rust.
113	Wire, wiring methods not as specified.
114	Mounting characteristics not as specified.
115	O-rings missing; damaged; not lubricated as required.
116	Identification/information plates missing, damaged or nonconforming; not secured and positioned as required.
Minor	
201	Preservation, packaging, and packing not as specified.
202	Marking and manufacturer's identification, missing, incorrect, illegible, not permanent, or not located as specified.

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4.5.1.3 Sampling for group B inspection. As a minimum, the contractor shall randomly select a sample quantity from each lot of completed enclosed switches in accordance with table VI and inspect them in accordance with table V in the order shown. The sample quantity shall be selected from inspection lots that have passed group A inspection. If one or more defects are found in any sample, the entire lot represented by the sample shall be rejected. If a lot is rejected, the contractor has the option of screening 100 percent of the lot for the defective characteristic(s) or providing a new lot which shall be inspected in accordance with the sampling plan contained herein. The contractor shall maintain for a period of 3 years after contract completion all records of inspections, tests, and any resulting rejections.

TABLE V. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph
Level of effectiveness of enclosure	3.6.6	4.7.7
Insulation resistance (where applicable)	3.6.2	4.7.3
Dielectric withstanding voltage (where applicable)	3.6.1	4.7.2

TABLE VI. Sampling for group B inspection.

Lot size	Sample size
2 to 25	3
26 to 50	5
51 to 90	6
91 to 150	7
151 to 280	10
281 to 500	11
501 to 1,200	15
1,201 to 3,200	18
3,201 to 10,000	22

4.6 Packaging inspection. Sample packages and packs and the inspection of the packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

4.7 Methods of inspection.

4.7.1 Visual and dimensional examination. Enclosed switches shall be visually examined to verify that the materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements (see 3.1, 3.3, 3.4, 3.5, 3.7, 3.8 and 3.9).

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4.7.2 Dielectric withstanding voltage. Enclosed switches shall be subjected to a dielectric withstanding voltage test in accordance with method 301 of MIL-STD-202. The following details shall apply:

(a) Test voltage: twice rated voltage plus 1000 V.
(b) Nature of potential: a.c. (rms).
(c) Points of application: between current carrying parts and enclosure (ground). Switches shall be in the energized (closed) position.
(d) Acceptance criteria: conformance to the requirements of 3.6.1.

4.7.3 Insulation resistance. Insulation resistance shall be measured in accordance with method 302, test condition B of MIL-STD-202. The test voltage shall be applied between current carrying parts of opposite polarity and live parts and the enclosure (ground). Switches shall be in the energized (closed) position. The enclosed switch shall conform to the requirements of 3.6.2.

4.7.4 Mechanical operation. Mechanical operation shall consist of the following cycle, repeated three times:

(a) The cover shall be completely removed and reassembled on the enclosure.
(b) The switch operating shaft shall be rotated through all positions three times.

The enclosed switch shall conform to the requirements of 3.6.3.

4.7.5 Vibration. Enclosed switches shall be subjected to type I vibration tests of MIL-STD-167-1. The following details and exceptions shall apply:

(a) Frequency - The variable frequency tests shall be omitted.
(b) Electrical test conditions - The test circuits shall be circuit B in accordance with method 310 of MIL-STD-202, and shall consist of one pair of closed mated contacts for each six sections. At the option of the manufacturer, contacts may be wired in series.
(c) Examination after test - Enclosed switches shall be checked for proper operation and examined for change in switch shaft position, and evidence of broken, loose, deformed, or displaced parts. Defective lamps or fuses shall not be cause for rejection.

4.7.6 Shock. Enclosed switches shall be subjected to the high-impact shock test for grade A, type A, class I equipment in accordance with MIL-S-901. The conditions of 4.7.5(b) and (c) shall apply. After completion of the test, where applicable (see 3.1), electrical indicator circuits shall be fitted with lamps and fuses, as required, and energized at rated voltage. Failure of the lamps to illuminate shall be cause for rejection. Failure to satisfactorily pass the level of effectiveness or dielectric withstanding voltage tests (where applicable) shall be cause for rejection.

4.7.7 Level of effectiveness of enclosure. Enclosed switches shall be tested in accordance with MIL-STD-108 except they shall not be energized. The degree of enclosure shall be as specified (see 3.1). Acceptance/rejection criteria shall be as specified in MIL-STD-108.

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4.7.8 Salt spray. Enclosed switches shall be tested in accordance with method 101, test condition A of MIL-STD-202. Evidence of excessive corrosion as specified in 3.6.7 shall be cause for rejection.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisitions. For the intent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.4.)

5.1 Preservation - packaging, packing and marking. Enclosed switches shall be individually preserved-packaged level A or C packed level A, B, or C as specified (see 6.2) and marked in accordance with MIL-E-17555. Special marking shall be as specified (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Enclosed rotary switches covered by this specification are intended for use in electrical and electronic power and control circuits in exposed locations on Naval ships.

6.2 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification, and applicable specification sheet and Military part number/symbol required.
(b) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(c) When first article testing is required (see 3.2).
(d) If fuses are required (see 3.4.5).
(e) If lamps are required (see 3.4.11).
(f) If holes for stuffing tubes or cable entrance is required (see location 3.5.1.1).
(g) If stuffing tubes, packing, and holes for stuffing tubes are required (see 3.5.1.1).
(h) If painting is to be other than specified (see 3.7.3).
(i) Data for information plates, if required (see 3.8.1.2).
(j) Level of preservation, packaging and packing required (see 5.1).
(k) If special marking is required (see 5.1).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

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Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.4	DI-NDTI-80809	Test, reports	----

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first ___ production items, a standard production item from the contractor's current inventory (see 3.2), and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4.1 First article extension. Part number listed in reverse order of acceptability. Part numbers not listed below are not included in extension coverage.

M15743/x	Current (amps)
1-001	10
2-001	10
3-001/2/4	10
16-001	10
17-001	10
18-001	10
19-001	10
20-001	10
3-003	10
4-002	10
6-001	10
4-001	10
5-001	10
5-002/3	10
8-001	30
8-003	
8-002	
9-001	
9-003	
9-002-11-001	

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6.5 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.6 Symbol number. The symbol number is a standard electrical equipment designation. Symbol numbers are listed in DOD-HDBK-290.

6.7 Superseded drawings. The superseded drawings are listed on the applicable specification sheet (see 3.1).

6.8 Subject term (key word) listing.

Cables
Fuses
Shaft extensions
Stuffing tubes
Terminal assemblies
Washers

6.9 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 5930-N746)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-S-15743C(SH)	22 NOVEMBER 1993

3. DOCUMENT TITLE
SWITCHES, ROTARY, ENCLOSED

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
COMMANDER NAVAL SEA SYSTEMS COMMAND (SEA 03R42)	(1) Commercial (703) 332-9137	(2) AUTOVON (AV 332-9137
c. ADDRESS (Include Zip Code) 2531 JEFFERSON DAVIS HWY ARLINGTON, VA 22242-5160	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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